

HAMDAN TARIQ

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EDUCATION

National University of Sciences & Technology (NUST) <i>Bachelor of Science in Artificial Intelligence</i>	Quetta, Pakistan 2024 - 2028
• Majors: Machine Learning, Programming for AI, Data Structures & Algorithms, Database Systems, Cloud Computing, Digital Logic Design, Artificial Intelligence • GPA: 3.79/4.00 (Transcript) • Credits earned: 51	
Tameer-i-Nau Public College <i>HSSC Pre-Engineering</i>	Quetta, Pakistan 2022 - 2024
• Percentage: 88.5%	

RESEARCH PROJECTS

Gesture-Controlled Robotic Hand <i>Embedded System</i> <i>National University of Sciences & Technology (GitHub)</i>	Feb 2025 - June 2025 <i>DLD Project</i>
• Built a gesture-controlled robotic hand using MPU6050 gloves for human-machine interaction • Captured hand orientation via gyroscope and accelerometer for precise directional control • Mapped glove gestures to robotic actions with button-triggered grip open/close mechanism • Applied embedded systems and sensor integration to deliver a functional real-time control system	
Brain Tumor Segmentation using U-Net <i>Python, AI</i> <i>National University of Sciences & Technology</i>	Feb 2026 - May 2026 <i>ML Project</i>
• Built a 2D U-Net with custom encoder-decoder on BraTS 2020 for brain tumor segmentation • Designed multi-level feature channels (96, 192, 384) to capture spatial context across scales • Enhanced generalization via mask generation and augmentation strategies	
Animal Detection System <i>Computer Vision, Deep Learning</i> <i>National University of Sciences & Technology</i>	Feb 2026 - May 2026 <i>PFA Project</i>
• Built video-based animal detection system auto-annotating 52 classes using RT-DETR & YOLO • Leveraged RT-DETR for high-resolution and YOLO for fast-moving object detection • Fused both models via Mixture of Experts for robust performance across diverse video conditions	
LAN Chat & File Transfer App <i>C++, Systems Design</i> <i>Independent Project (GitHub)</i>	Sep 2025 - Jan 2026 <i>CN Project</i>
• Built a C++ LAN chat app with Qt GUI and custom async non-blocking UDP network • Implemented auto peer discovery and chunk-based file transfer without manual IP configuration • Delivered real-time presence detection demonstrating expertise in networking, and system design	
Character Recognition CNN <i>Computer Vision</i> <i>Independent Project (GitHub)</i>	Mar 2026
• Built a custom CNN for alphanumeric character recognition from document images • Trained on labeled datasets achieving accurate character classification across digits and letters • Gained practical expertise in CNNs, image preprocessing, and deep learning training workflows	

VOLUNTEER EXPERIENCE

Summer Intern

Al-Khidmat Foundation Hospital

Supervisor: Muhammad Sannan Akbar

June 2025 - July 2025

Quetta, Balochistan, Pakistan

Total Hours: 48

- Completed volunteer internship covering leadership, entrepreneurship, community development
- Contributed to environmental fundraising initiative securing donations for tree plantation
- Gained hands-on experience in teamwork, project coordination, and professional report writing

TECHNICAL SKILLS

Programming: Python, C++, System Verilog, Assembly Language, HTML, CSS

Tools: Google Colab, PyCharm, Visual Studio, Ubuntu, Microsoft Office, LaTeX, Kali Linux, SQL, Streamlit, QT Creator, Arduino IDE

Libraries: Scikit-Learn, TensorFlow, PyTorch, OpenCV, Pandas, Numpy, Matplotlib, Seaborn, Keras, ASIO (C++), OpenGL

SOFT SKILLS

Problem Solving

Project & Time Management

Leadership & Teamwork

REFERENCES

Name	Organization	Title	Phone	Email
Mohsin Raza Jafri	NUST, NBC, Quetta	Dr.	+92-333-6352864	mjafari@nbc.nust.edu.pk
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